

Yuming Jin

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RESEARCH INTERESTS

Global Climate Change; Carbon and Oxygen Cycle; Large-scale Atmospheric Dynamics; Carbon-Climate Feedback; Air-Sea interaction

EMPLOYMENT

Postdoctoral Fellow	2026 - present
University of California, Santa Barbara, Earth Research Institute	CA, USA
Advanced Study Program (ASP) Postdoctoral Fellow	2024 – 2026
National Center for Atmospheric Research (NCAR), Earth Observing Laboratory & Climate and Global Dynamics	CO, USA
Graduate Student Researcher	2018-2023
Scripps Institution of Oceanography, University of California San Diego	CA, USA

EDUCATION

Ph.D. Earth Sciences (Climate, Atmospheric Sciences and Physical Oceanography)	2018 - 2023
Scripps Institution of Oceanography, University of California San Diego	CA, USA
Thesis: <i>Advances in Representing Atmospheric Circulation and Structure for Carbon Cycle Studies</i> (Awarded UCSD Chancellor's Dissertation Medal)	
Advisor: <i>Dr. Ralph F. Keeling</i>	
B.S. Environmental Sciences	2014 - 2018
Donghua University	Shanghai, China
Graduation with honor: College Graduate Excellence Award of Shanghai	
Thesis: <i>Preparation and application of Bi₂MoO₆-based recyclable visible-light driven photocatalyst</i> (Awarded University Best Undergraduate Thesis)	
Advisor: <i>Dr. Lisha Zhang</i>	

FUNDING & AWARDS

Fellowships and Scholarships

NCAR ASP Postdoc Fellowship (\$240,000)	2024-2026
SIO Department Graduate Student Research Award (\$3,000)	2023

Henry & Grace Doherty Fellowship Fund (1 year stipend and tuition)	2018-2019
Hariko Quirk Endowment Fund (1 year stipend and tuition)	2018-2019
National Scholarship, Ministry of Education of China (~\$3,700)	2015&2016&2017
First Class Scholarship & Outstanding Student Award, Donghua University (~\$1300)	2015&2016&2017

Awards

UCSD Chancellor's Dissertation Medal	2024
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LIST OF PUBLICATIONS [[Google Scholar](#); [ResearchGate](#)]

MANUSCRIPTS UNDER REVIEW

14. **Jin, Y.**, Morgan, E., Stephens, B., Landschützer, P., DeVries, T.: Ocean heat-oxygen-carbon coupling constrains the climate-driven carbon sink, **Under review at Global Biogeochem. Cy.**
13. Morgan, E., **Jin, Y.**, Tohjima, Y., Keeling, R.: Atmospheric O₂ measurements show a strengthening ocean carbon sink and a stagnate terrestrial carbon sink since 1990, **Under review at Science**
12. **Jin, Y.**, Stephens, B., Morgan, E., Manizza, M., Sharp, J.: Global Gridded Air-sea O₂ Flux Inferred from Machine Learning-based Dissolved Oxygen Products, **Under review at ESSD**

PEER-REVIEWED PUBLICATIONS

11. Stephens, B., **Jin, Y.**, Sweeney, C., McKain, K., Gaubert, B., Baker, D., Basu, S., Bertolacci, M., Chevallier, F., Commane, R., Crowell, S., Deng, F., Johnson, M., Keeling, R., Liu, J., Liu, Z., Maity, S., Morgan, E., Patra, P., Philip, S., Wofsy, S., Zammit-Mangio, A., Improved latitudinal carbon budgets from global airborne surveys, *PNAS*, 2026, 125(25), e2523984123, <https://doi.org/10.1073/pnas.2523984123>
Highlights: [NCAR News](#)
10. **Jin, Y.**, Stephens, B., Long, M., Manizza M., Lovenduski, N., Nevison, C., Morgan, E., Keeling, R.: Atmospheric Oxygen Constraints on Southern Ocean Productivity and Drivers of Carbon Uptake, *Nat. Geosci.*, 2026, 19 (5), <https://doi.org/10.1038/s41561-026-01944-z>
Highlights: [Nature Geoscience News & Views](#), [NCAR News](#), [earth.com](#)
9. **Jin, Y.**, Stephens, B., Long, M., Chandra, N., Chevallier, F., Hooghiem, J., Luijkx, I., Maksyutov, S., Morgan, E., Niwa, Y., Patra, P., Rödenbeck, C., Vance J.: The Atmospheric Potential Oxygen forward Model Intercomparison Project (APO-MIP): Evaluating simulated atmospheric transport of air-sea gas exchange tracers and APO flux products, *Geosci. Model Dev.*, 2025, 18, 5937–5969, <https://doi.org/10.5194/gmd-18-5937-2025>

8. Liu, Z., Rogers, B., Keppel-Aleks, G., Helbig, M., Ballantyne, A., Kimball, J., Chatterjee, A., Foster, A., Kaushik, A., Virkkala, A., Burrell, A., Schwalm, C., Sweeney, C., Schuur, E., Dean, J., Watts, J., Kim, J., Wang, J., Hu, L., Welp, L., Berner, L., Mauritz, M., Farina, M., Mack, M., Parazoo, N., Madani, N., Keeling, R., Commane, R., Goetz, S., Piao, S., Natali, S., Wang, W., Buermann, W., Walker, X., Lin, X., Wang, X., **Jin, Y.**, Yu, K., Zhang, Y.: Drivers and trends of enhanced seasonal CO₂ amplitude in northern high latitudes, *Nat. Rev. Earth & Env.*, 2024, <https://doi.org/10.1038/s43017-024-00600-7>
7. **Jin, Y.**, Keeling, R. F., Long, M. C., Stephens, B. B., Patra, P., Rödenbeck, C., Morgan, E. J., Kort, E. A., Sweeney, C.: Improved Atmospheric Constraint of the Southern Ocean CO₂ Exchange, *PNAS*, 2024, 121(6), e2309333121, <https://doi.org/10.1073/pnas.2309333121>
6. **Jin, Y.**, Stephens, B. B., Keeling, R. F., Morgan, E. J., Rödenbeck, C., Patra, P., Long, M. C.: Seasonal Tropospheric Distribution and Air-sea Fluxes of Atmospheric Potential Oxygen from Global Airborne Observations, *Global Biogeochem. Cy.*, 2023, 37, e2023GB007827, <https://doi.org/10.1029/2023GB007827>
5. **Jin, Y.**, Keeling, R. F., Rödenbeck, C., Patra, P., Piper, S., and Schwarzman, A.: Impact of Changing Winds on the Mauna Loa CO₂ Seasonal Cycle in Relation to the Pacific Decadal Oscillation, *JGR-Atmosphere*, 2022, 127(13), <https://doi.org/10.1029/2021JD035892>
4. Morgan, E., Manizza, Manfredi., Keeling, R. F., Resplandy, L., Mikaloff-Fletcher, S.E., Nevison, C.D., **Jin, Y.**, Bent, J.D., Aumont, O., Doney, S.C., Dunne, J.P., John, J., Lima, I.D., Long, M.C., Rodgers, K.B.: An atmospheric constraint on the seasonal air–sea exchange of oxygen and heat in the extratropics, *JGR-Ocean*, 2021, 126(8), e2021JC017510, <https://doi.org/10.1029/2021JC017510>
3. **Jin, Y.**, Keeling, R. F., Morgan, E. J., Ray, E., Parazoo, N. C., and Stephens, B. B.: A mass-weighted isentropic coordinate for mapping chemical tracers and computing atmospheric inventories, *Atmos. Chem. Phys.*, 2021, 21, 217–238, <https://doi.org/10.5194/acp-21-217-2021>
2. **Jin, Y.**, Shen, X., Liu, Z., Wang, Z., Zhu, B., Xu, P., Luo, L., Zhang, L.: Synthesis of NiTiO₃–Bi₂MoO₆ core–shell fiber-shaped heterojunctions as efficient and easily recyclable photocatalysts. *New J. Chem.* 2018; 42(1): 411-9, <https://doi.org/10.1039/C7NJ03367B>
1. Lou, X., Fang, C., Geng, Z., **Jin, Y.**, Xiao, D., Wang, Z., Liu, J., Guo, Y.: Significantly enhanced base activation of peroxymonosulfate by polyphosphates: Kinetics and mechanism. *Chemosphere*. 2017; 173: 529-34, <https://doi.org/10.1016/j.chemosphere.2017.01.093>

OTHER PUBLICATIONS

1. Betts, R., Jones, C., **Jin, Y.**, Keeling, R. F., Kennedy, J. J., Knight, J. R.: Analysis: What impact will the coronavirus pandemic have on atmospheric CO₂, Carbon Brief, 2020

DATASETS & MODELS

2. **Jin, Y.:** GOBAI-O₂-Flux v1.0: Global Gridded Air-sea Oxygen Flux Inferred from a Machine Learning-based Dissolved Oxygen Product, Zenodo, 2025, <https://doi.org/10.5281/zenodo.17765275>
1. Stephens, B., Long, M., **Jin, Y.**, Chandra, N., Chevallier, F., Hooghiem, J., Luijkx, I., Maksyutov, S., Morgan, E., Niwa, Y., Patra, P., Rödenbeck, C., Vance, J.: Atmospheric Potential Oxygen forward Model Intercomparison Project (APO-MIP), NCAR Research Data Archive, 2025, <https://doi.org/10.5065/F3PW-A676>

TEACHING

Guest Lectures

Graduate Marine Science Speaker Seminar Series, UC Santa Barbara

2026 Spring

Teaching Assistant

Physical Basis of Global Warming, UC San Diego

2023 Fall

Tutor

Math Bootcamp, Scripps Institution of Oceanography

2019 Summer

SEMINAR (INVITED) TALKS

7. Atmospheric Constraints on Southern Ocean Carbon Cycle, UCSB IGPMS Seminar, 21 April, 2026
6. Constraining Southern Ocean Seasonal CO₂ Uptake with Atmospheric O₂ and CO₂ Measurements, SOFLUX Webinar, 6 June, 2024
5. Improving estimates of carbon sources, sinks, and atmospheric transport using global airborne data, JPL Carbon Club Seminar, 18 April, 2024
4. Using airborne CO₂ observations and a mass-indexed isentropic coordinate to test transport models and improve surface flux estimates, OCO₂-Flux telecon, 23 January, 2024
3. Quantification of seasonal air-sea CO₂ and O₂ exchange at regional to global scale using data from airborne campaigns, Shanghai Jiao Tong University, School of Oceanography Seminar, 6 April, 2023
2. Atmospheric CO₂ and O₂ observations as indicators of global climate and biogeochemistry change, Carbon Monitor Seminar, 24 July, 2022
1. Improved Atmospheric Constraint of the Southern Ocean CO₂ Sink, BGC-Argo Science Meeting, 18 May, 2022

CONFERENCE & WORKSHOP PRESENTATIONS

16. **Jin Y.:** Atmospheric O₂ Constraints on Ocean Biogeochemistry and Physics, CLIVAR Oxygen Workshop, Beijing, 23-26 Oct., 2025 (Oral)

15. **Jin Y.:** Atmospheric constraints on the Southern Ocean productivity and seasonal air-sea CO₂ and O₂ exchange, CLIVAR Summit 2025, Boulder, 21-23 July, 2025 (Oral)
14. **Jin Y.:** An Isentrope-Based Framework for Diagnosing Tracer Transport and Surface Fluxes, TransCom Workshop, Online, 28 May, 2025 (Oral)
13. **Jin, Y.:** Improving estimates of carbon sources, sinks, and atmospheric transport using global airborne data and a transformed isentrope coordinate, ICDC 2024, Manaus, 29 July - 2 Aug., 2024 (Oral)
12. **Jin, Y.:** Refining transport models: leveraging airborne tracer observations and cross-isentrope diabatic mixing metrics, TransCom-24, Boulder, 28 May, 2024 (Oral)
11. **Jin, Y.:** Southern Ocean Seasonal O₂ and CO₂ Exchange Inferred from Airborne Observations, Ocean Sciences Meeting 2024, New Orleans, 18-23 Feb., 2024 (Poster)
10. **Jin, Y. et al.:** Long-term and decadal changes of Mauna Loa CO₂ seasonal cycle in relation to wind shifts driven by Pacific Decadal Oscillation, AGU Fall Meeting 2023, San Francisco, 11-15 Dec., 2023 (Poster)
9. **Jin, Y. et al.:** Seasonal Air-sea Atmospheric Potential Oxygen Flux Inferred from Global Airborne Observations, 4th Workshop on Atmospheric Oxygen, Brunswick, 23-25 Aug., 2023 (Oral)
8. **Jin, Y. et al.:** Using airborne CO₂ observations and a mass-indexed isentropic coordinate to test transport models and improve surface flux estimates, NOAA GML Global Monitoring Annual Conference 2023, Boulder, 23-24 May, 2023 (Poster)
7. **Jin, Y. et al.:** Constraining Atmospheric Transport Models and Improving Surface CO₂ Flux Estimates Using Airborne CO₂ Observations and a Mass-Indexed Isentropic Coordinate, AMS 2023, Denver, CO, 9 Jan. – 12 Jan., 2023 (Oral)
6. **Jin, Y. et al.:** Using airborne CO₂ observations and a mass-indexed isentropic coordinate to test transport models and improve surface flux estimates, TransCom Meeting, Wageningen, Netherlands, 16-17 Sept., 2022 (Oral)
5. **Jin, Y. et al.:** Constraining the Seasonal Cycle of the Southern Ocean CO₂ Uptake with Airborne Observations, Ocean Sciences Meeting 2022, Virtual Meeting, 28 Feb. – 4 Mar., 2022 (Oral)
4. **Jin, Y. et al.:** Constraining the Seasonal Cycle of the Southern Ocean CO₂ Uptake with Airborne Observations, AGU Fall Meeting 2021, New Orleans, 13-18 Dec., 2021 (Oral)
3. **Jin, Y. et al.:** Mapping Atmospheric CO₂ and O₂ from Airborne Observations Using a Mass-weighted Isentropic Coordinate, NOAA GML Virtual Global Monitoring Annual Conference 2021, Virtual Meeting, 24-28 May, 2021 (Poster)

2. **Jin, Y.** et al.: Mapping the CO₂ Seasonal Cycle from Airborne Observations Using a Mass-Weighted Isentropic Coordinate, American Meteorology Society Annual Meeting, Virtual Meeting, 9-15 Jan., 2021 (Oral)
1. **Jin, Y.** et al.: Precise quantification of seasonal air-sea exchanges of O₂ at the hemispheric scale using data from the ATom, ORCAS, and HIPPO airborne campaigns, Ocean Sciences Meeting 2020, San Diego, CA, 16-21 Feb., 2020 (Poster)

SERVICE & OUTREACH

Reviewer: AGU Advances, Atmospheric Chemistry and Physics, Biogeosciences, Earth and Space Science, Earth System Science Data, Environmental Research Communications, Environmental Research Letters, Global Biogeochemical Cycles, Journal of Advances in Modeling Earth Systems, Journal of Geophysical Research-Atmosphere, Journal of Geophysical Research-Ocean, Nature Communication, Science Advances, Scientific Reports

Committee Member:

ASP Seminar Series Committee (2023-2025)

AI4Carbon Committee Organizer (2024-present)

Judge: Scripps Student Symposium (2021)

Conference Organizer/Convener:

NCAR Distinguished Lecture Series, 16-17 April 2024, Boulder, USA

NCAR Expanding Horizon Seminar series, 19 September 2024, Boulder, USA

EGU2025: Understanding feedbacks between greenhouse gas exchange processes and climate variability using in situ observations, remote sensing, and machine learning, 27 April – 2 May 2025, Vienna, Austria

WORKSHOPS AND TRAININGS

GO-BGC Workshop (Aug. 2026)

Workshop for New Advances in Land Carbon Cycle Modeling (June 2026)

Summer School for Inverse Modeling of Greenhouse Gases Summer School (June 2024)

CESM Tutorial (Aug. 2020)

TECHNICAL SKILLS

Programming Language: R (expert), Python (intermediate), Fortran (intermediate), Julia (advanced beginner), Bash

Modeling Experience: CESM, GEOS-Chem, TM3, HYSPLIT, Jena CarboScope Inversion